



Today I am learning to refine my explanations and critique chance situations for fairness and reason.

1. Critique the Situation

Read each situation. Decide if it is fair or unfair. Explain your reasoning.

Situation	Fair or Unfair?	Why?
a) A game: Spin a spinner with 3 red sections and 1 blue section. The winner gets a prize.	_____	_____
b) A raffle: 5 tickets are sold. The organiser has 2 tickets, too.	_____	_____
c) A coin toss decides who goes first in a game.	_____	_____
d) A class vote: The teacher counts some votes as 2 because they are from prefects.	_____	_____

2. Analyse and Explain

Study the data and answer the questions.

A website asked 400 people about their favourite type of pet.

Pet Type	Number of People
Dog	176
Cat	128
Fish	48
Bird	24
Reptile	24



- What percentage of people preferred dogs? _____ %
- If one person is chosen at random from this group, what is the probability they prefer a cat? Write as a fraction and as a percentage. _____ %
- Is it more likely to choose someone who prefers a bird or a reptile? Why? _____
- The website claims "Most people love pets that are not cats." Is this claim supported by the data? Explain. _____

3. Fairness Investigation

A game uses a number cube (1–6) and these rules.

- Roll 1 or 2 → Player A scores a point.
- Roll 3, 4 or 5 → Player B scores a point.
- Roll 6 → No one scores.



- Find the probability of each player scoring a point.
Player A: _____ Player B: _____
- Is this game fair? Explain your reasoning.

- Change the rules to make the game fair. Write your new rules clearly.

4. Real-Life Critique

Read the situation and answer the questions.

A company offers a "lucky draw" with these chances of winning.



Prize	Number of Sections
\$100	1
\$50	1
\$20	1
\$10	1
Nothing	2
Total	6

- What is the probability of winning \$100? _____
- What is the probability of winning nothing? _____
- Is this a fair draw? Explain your answer. _____
- The company says "You have a great chance to win a prize!" Do you agree? Use the probabilities to support your opinion. _____

5. Compare and Justify

Two bags are used in a game. You pick one counter without looking.

Bag A	Colour	Number	Bag B	Colour	Number
	Red	4		Red	3
	Blue	6		Green	7
	Total	10		Total	10

- What is the probability of picking a red counter from each bag?
Bag A: _____ Bag B: _____
- Which bag has a greater chance of picking a red counter? Explain.

- Which bag has a greater chance of picking a blue or green counter? Explain.

6. Explanation Challenge

Answer the questions using complete sentences and mathematical reasoning.

- Why is it important to consider both probability and fairness in games? Give an example.

- A friend says, "I lost 5 times in a row, so I'm more likely to win next time." Explain why this is a misunderstanding.

- How can understanding probability help us make fairer decisions in real life?



7. Design a Fair Game

Design your own game that is fair.

- Include:
- Rules
 - What is needed
 - Why it is fair (use probabilities)



My Game:

8. Reflect

Tick (✓) the box that shows how confident you feel.

I can...	Not yet	Getting there	Confident
critique chance situations for fairness.	☐	☐	☐
explain probabilities using fractions, decimals or percentages.	☐	☐	☐
use data to explain and justify my conclusions.	☐	☐	☐
apply probability to real-life contexts.	☐	☐	☐



I can...

- ☐ critique chance situations.
- ☐ explain and justify using probability.
- ☐ evaluate fairness in games and data.
- ☐ apply probability in real-life contexts.
- ☐ make fair and informed decisions.

Chance Words



Great job!

You are refining your thinking and using mathematics to make fair and smart choices!